EXPLORING THE IMPACT OF DATA DASHBOARDS ON FORMULA 1 ENGAGEMENT INTENTIONS AS MEASURED BY THE SPORTS FAN MODEL OF GOAL-DIRECTED BEHAVIOUR

A study submitted in partial fulfilment
of the requirements for the degree of
MSc Data Science

at

THE UNIVERSITY OF SHEFFIELD

by

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Word Count: 10,731

August 2025

Abstract

Background: Existing research into the applicability of the Sports-Fan Model of Goal-Directed Behaviour is extremely scarce, particularly in the context of Formula 1. Similarly, there is a lack of research into the potential value of using data dashboards to boost Formula 1 engagement.

Aims: The present study aimed to evaluate the Sports-Fan Model of Goal-Directed Behaviour and examine how data dashboards might impact intentions to engage with Formula 1 for both fans and non-fans.

Methods: A total of 48 participants took part in this study involving an independent groups design where one group was shown a Formula 1 video, a related data dashboard, and a questionnaire, whilst the other group was given only the video and the questionnaire. The study utilised a quantitative approach involving regression analyses, mediation analyses, and Two-Way ANCOVAs.

Results: The only significant paths of the model were from Perceived Behavioural Control to Desire, from Fan Engagement to Intention, and from Past Satisfaction to Intention. The model was found to explain 88.9% of the variance in intention, but this was assumed to be an inflated value due to overfitting. The Two-Way ANCOVA revealed that the dashboard group had greater intentions to engage with Formula 1 than the non-dashboard group, and this effect remained after controlling for fan-status.

Conclusions: It was concluded that the Sports-Fan Model is likely overcomplicated compared with the standard Model of Goal-Directed Behaviour, and it does not significantly improve predictive power. It was also concluded that data dashboards have the potential to increase Formula 1 engagement regardless of fandom. However, due to various methodological limitations, further research in this area is recommended.

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Introduction

According to a YouGov survey, around 35% of Britons watch sports at least once per week, while only 17% claim to never watch sports (Dinic, 2024). Live sports constitute a £1.7 Billion market in the UK and is projected to grow by over 20% by 2029 (Walmsley, 2024). Globally, the sport streaming market is expected to reach $134 billion by 2030 as access and interest in sports increase (Shantanu, 2022). One of the fastest-growing sports in recent years is The FIA Formula One World Championship, more commonly known as Formula 1 (F1). This is the highest class of open-wheel single-seater racing, where typically 20 drivers across 10 teams compete in a series of races, or Grands Prix, throughout a season that takes place within a single calendar year. In F1, points are awarded according to drivers' finishing positions. The 10 highest-placing drivers in each race are awarded points ranging from 1 point for 10th place to 25 points for 1st place. In 2021, so-called "sprint races" were introduced to the F1 season as a way of enhancing the racing spectacle and capitalising on F1's rise in popularity. These races are shorter than ordinary Grands Prix, award fewer points, and are only incorporated at a select few locations throughout the season. At the end of each season, the Drivers' Championship is awarded to the driver who accumulated the most points throughout the season, whilst the Constructors' Championship is awarded to the team whose drivers obtained the most combined points.

The sport has as many as 750 million fans worldwide (Brown, 2024), and a valuation of around $18 billion (Ozanian, 2024). However, maintaining growth and relevance in an increasingly saturated entertainment landscape is challenging. Noble (2023) highlighted that Netflix's Drive to Survive series and a controversial 2021 season saw a rapid growth in F1's popularity, but since then the number of social media mentions has dropped, from 6.14 million in 2022 to 1.83

million in 2023. This volatility highlights the importance of engagement strategies to protect against periods of stagnation.

Engagement in this context can be described as an individual's involvement with F1 through behaviours such as watching races, discussing the sport with others, or sharing related content on social media. The present study adopts a theory-based understanding of engagement that draws from the Sports Fan Model of Goal-Directed Behaviour (SFMGB; Yim & Byon, 2021). As such, engagement is conceptualised not merely as observable behaviour, but as an outcome of several measurable determinants. The benefit of operationalising engagement in this way is that it eliminates the need to observe or track the behaviour of participants, which can be a costly process. However, there is very little existing research into the validity of the SFMGB, particularly in the context of F1, and this is what the present study aims to address. This study also discusses the possibility that data dashboards be used as tools for current or prospective F1 viewers to enhance engagement, as measured by the SFMGB. It is also important to understand whether such a tool could be used for both fan retention and fan attraction. As such, the objectives of this research are as follows:

1. To evaluate the applicability of the SFMGB in the context of Formula 1.
2. To examine whether exposure to a Formula 1 data dashboard impacts intentions to engage with the sport.
3. To assess whether the effect of the data dashboard on intentions differs between fans and non-fans.

Literature Review

Benefits of Sports Engagement

Research suggests that sports viewership is associated with psychological well-being. Kim et al. (2017) investigated psychological factors of participants before and after watching a sports event and found improvements in measures of need-fulfilment and multiple measures of well-being. Furthermore, Kinoshita et al. (2024) utilised a multi-disciplinary approach and found that watching sport was associated with both self-reported well-being and neurophysiological indicators of well-being. One possible explanation for these benefits is that the social connections formed through collective viewership contribute to well-being (Wann, 2006). This theory is supported by Kinoshita et al. who found that well-being improvements were stronger for popular sports compared with less popular sports. However, it remains unclear whether sport popularity improves well-being via the collective viewership as suggested by Wann, or if the popularity of the sport is dependent upon the extent to which it provides a viewing experience that improves well-being. Another way in which well-being may be improved is through the emotional regulation of experiencing wins and losses. Cayolla (2024) used neuroimaging techniques to assess sports fans' reactions to positive and negative moments for their respective teams. It was found that the neurological structures relating to emotion and memory responded more strongly to positive events than negative events, suggesting that emotion regulation played an important role in dealing with the negative stimuli.

Watching sports may also encourage healthier lifestyles. Kawakami et al. (2024) assessed adults' frequency of watching sports, in the media or in-person, and measured various health indicators a year later. It was found that watching sports was associated with a reduced likelihood of becoming physically inactive or skipping breakfast. This suggests that watching sports may encourage people to maintain healthier lifestyles. It has also been suggested that the positive impacts of viewing sports start early in childhood, as Zhang (2022) found that watching sports-related cartoons in childhood indirectly promoted physical activity in adulthood. There is also evidence that attending football games encourages physical activity as venues often promote health initiatives (Parry et al., 2019). However, such evidence is mixed because Parry et al. found that the unhealthy food options at football games may counteract the positive impacts of health initiatives. Furthermore, Tsuji et al. (2021) found that watching sports on television was associated with an increased risk of obesity, and attending sports events in person at least once per week was associated with increased alcoholism and aggression. The results of these studies should be interpreted cautiously as it is difficult to ascertain causation. It is logical that a general interest in sport may be the influencing factor in both watching sports and remaining physically active, so it is unclear whether watching sports encourages physical activity directly. Furthermore, whilst watching sports on television is associated with obesity, this may be a result of pre-existing mobility difficulties faced by those with obesity which prevents them from watching sports in person.

Watching sports has also been associated with broader measures of life satisfaction and meaning. For example, Wann et al. (2017) investigated the relationship between team identification and meaning in life and found that sense of belonging mediated this relationship. This suggests that watching sports and supporting a sports team may be instrumental in facilitating a sense of belonging and meaning. A similar conclusion was drawn by Inoue et al. (2017), who found that increased team identification through watching live sports was associated with greater perceived emotional support which, in turn, improved life satisfaction. These studies suggest that the sense of belonging and identity that is facilitated by being part of a sports' fanbase is instrumental in improving individuals' life satisfaction. However, it is unclear whether this applies to those who do not watch sports in person. Oh et al. (2020) found that watching sports via media did not have the same benefit for life satisfaction than watching sports in-person. It has been posited by Jiang et al. (2021) that the social platform provided by watching sports in-person is critical in enhancing the sense of belonging, which could explain why this format has a greater influence on life satisfaction. However, as noted by Kan and Xie (2024), those who can afford to watch sports in-person are typically of a higher socioeconomic status, and this may confound the observed life satisfaction effects.

Whilst the benefits of F1 viewership specifically does not have literary backing, some of these benefits are theoretically applicable to the sport, especially with recent efforts to foster inclusivity. The introduction of the #WeRaceAsOne initiative (Formula 1, 2020) and shifts in viewer demographics are indicative of a sport that provides a sense of belonging. F1 CEO Stephano Dominicali stated that the percentage of female F1 fans increased from 7% to 40% between 2017 and 2022 (Hamlin, 2022), indicating that demographics that once made up a small proportion of fans are now feeling increasingly welcomed within the fanbase. This sense of belonging is likely to have a positive impact on life satisfaction, and thus should be encouraged (Wann et al., 2017). However, the potential physical benefits of watching F1 are less convincing. F1 is a sport that involves intense physical demands, with drivers typically having 8-12% bodyfat and regularly withstanding extreme G-forces and cockpit temperatures of over 40 degrees Celsius (Jones, 2023). Despite this, the sport is not accessible in a way that promotes physical activity among viewers. Unlike other sports such as football or basketball, where spectators may be inspired to emulate the athletes' behaviour for a relatively low cost, F1 does not offer a realistic pathway to take part. Viewers may be inspired to take part in other forms of competitive racing such as karting, but this is also extremely costly and lacks accessibility as a result. As such, whilst F1 celebrates exceptional physical performance, it is limited in its ability to inspire similar behaviour.

Despite F1's global popularity, research on engagement within this sport remains limited. F1 presents a unique spectator experience due to its technical and strategic complexity and these factors may influence how and why individuals engage with the sport, and whether models of sports engagement apply in this context. The wide-ranging potential benefits of watching sports mean it is important to research engagement strategies which could encourage F1 viewership.

Data Dashboards in Sports

One potential way to increase engagement with sports is by providing relevant data-analytics and visualisation tools. Visuo-spatial displays are vital for human communication. They have their origins in ancient caveman drawings and have evolved into modern data visualisation techniques (Hegarty, 2011). The visual representation of complex data plays a crucial role in facilitating pattern recognition and understanding (Gelman, Pasarica, & Dodhia, 2002; Kastellec & Leoni, 2007). Data dashboards integrate multiple types of data visualisation into a cohesive environment, allowing users to explore the data with interactive functionalities such as the ability to adjust data filters and freedom to alter visualisation appearance. Improvements in the accessibility and functionality of data dashboards has been instrumental in promoting their widespread adoption (Sinha, 2024) within areas of business, research, and sports.

The use of data in sports - often referred to as sports analytics - has traditionally focused on gaining a competitive edge by using data to improve strategy and performance (Szymanski, 2020; Perin et al., 2018). However, in recent years, data visualisation in sports has expanded to include aiding viewers' understanding, enjoyment and engagement. Many professional sports including football (Horky & Pelka, 2016), baseball (Korengold, 2022), and motorsport (Cuellar, 2025), have utilised visualisations shown on live broadcasts to aid viewer enjoyment or understanding. There is also evidence that these features are effective at fostering engagement, as Cho et al. (2023) found that fans who were shown sport-related data visualisations reported increased social identity and interaction with the sport.

Despite the growing incorporation of data visualisation during broadcast, fan-accessible data dashboards remain underutilised. One possible reason for this is that using a data dashboard whilst watching a sports event may be distracting and is likely to take the viewer’s attention away from the broadcast. With the sports sponsorship industry worth over $93 million (Gough, 2023), it is in the best interests of sponsors and broadcasters that new features increase viewership rather than reducing it. However, engagement with sport extends beyond live events. A report by Deloitte (2020) found that 95% of fans engaged with their respective sports in the off-season, and those who engaged with in-depth analyses of their teams’ performance spent an average of 37% more than those who consumed general news only. This suggests that providing a data dashboard feature outside of the live events could be an effective way of encouraging engagement and boosting revenue. Some sporting bodies have responded to this research, with NFL offering a Pro membership that includes exclusive data dashboards and Premier League providing a “stats centre” that contains comparative statistics.

F1 is renowned for its data-driven and analytical qualities, but it does not provide an official fan-oriented data dashboard feature. With 81% of F1 fans consuming related content outside of the live races (Brittle, 2024), there is a demand for additional analytical content, but the sport's current offerings are limited. F1's live timing feature is only usable during a live race and lacks the detailed statistics and visualisations that could be provided by a data dashboard. Third-party apps and websites have attempted to fill this gap but they lack exposure and accessibility because they are not officially endorsed by the sport. Given F1's highly analytical reputation and the increasing importance of utilising data-driven engagement in other sports, it is essential to investigate the potential impact of an F1 data dashboard to establish whether it could be a valuable investment for the sport.

Psychological Models of Behaviour and Intention

Intention can be defined as "the specific desired outcome that guides behaviour" (Freeman & Turvey, 2012). Therefore, since engagement is a form of behaviour, one way of operationalizing engagement with a sport is by focusing on intentions. There has been extensive research into the factors that contribute towards intentions, primarily through psychological models of behaviour.

The Theory of Planned Behaviour (TPB; Ajzen, 1991) is a psychological model of behaviour, based on the Theory of Reasoned Action (TRA; Ajzen & Fishbein, 1975) which describes how the performance of a behaviour is a result of intention, and states that there are three factors which contribute to intention. These factors are Attitudes, Subjective Norms (SN), and Perceived Behavioural Control (PBC). The TPB states that, collectively, these factors influence an individual's intentions to carry out a behaviour which, in turn, affects performance of the behaviour. Whilst there is evidence that all three determinants are correlated with behavioural intentions ($r \sim .70$; E.g., Godin & Kok 1996; Sheeran & Taylor 1999), it may not be a sufficient model. A meta-analysis by McEachan et al. (2011) found that the TPB explained 44.3% of the variance in intention, and 19.3% of the variance in behaviour. This means there is a significant proportion of both intentions and behaviour which remain unexplained by the model. There is also a distinct lack of research into the applicability of the TPB for sports engagement. Instead, the sport-related studies utilising this model tend to focus on sports participation and exercise (e.g., Norman et al., 2000; Rhodes and de Bruijn, 2013).

Nevertheless, the TRA and TPB have formed the basis for many models that have followed. A more comprehensive model was created by Perugini & Bagozzi (2001), who proposed an expansion to the TPB that incorporates additional determinants including desires, anticipated emotions, and the frequency and recency of past behaviours. This model is known as the Model of Goal-Directed Behaviour (MGB), and it addresses some elements that were omitted in the TPB. For example, including the frequency and recency of behaviour is relevant as both are indicative of future behaviour (Karg et al., 2021). Furthermore, the inclusion of anticipated emotional factors is vital as these have been shown to be important considerations in decision-making (van Kleef et al., 2021). Choe et al. (2019) assessed how the MGB applied to sports event attendance and found that anticipated emotions were a significant predictor of desire and, in turn, intentions. In fact, these attributes were stronger indicators of desire than subjective norms, which highlights the necessity of their inclusion within the model. Anticipated emotions may be especially relevant for predicting sports-viewing intentions because sports can produce emotional responses due to the individuals' identification with their respective team, athlete or nation. Overall, Choe et al. (2019) found the MGB to be a valid model for predicting the desire and intention to attend sporting events, explaining 52.7% of the variance in intention. However, they also acknowledged that the MGB may be insufficient in this context as it does not contain any determinants that are unique or specific to sports.

In response to the lack of sports-specificity provided by the MGB, a further expansion was created by Yim and Byon (2021) known as the Sports-Fan MGB (SFMGB). The SFMGB states that there are ten determinants of behavioural intention, which in-turn impact engagement with the sport. This model is shown in Figure 1, and it includes most of the attributes contained within the MGB, whilst addressing the need for additional sport-specific determinants including fan community identity, team identification, and fan engagement. A summary of each determinant is provided in Table 1.

Figure 1. *The Sports-Fan Model of Goal-Directed Behaviour (Yim & Byon, 2021)*

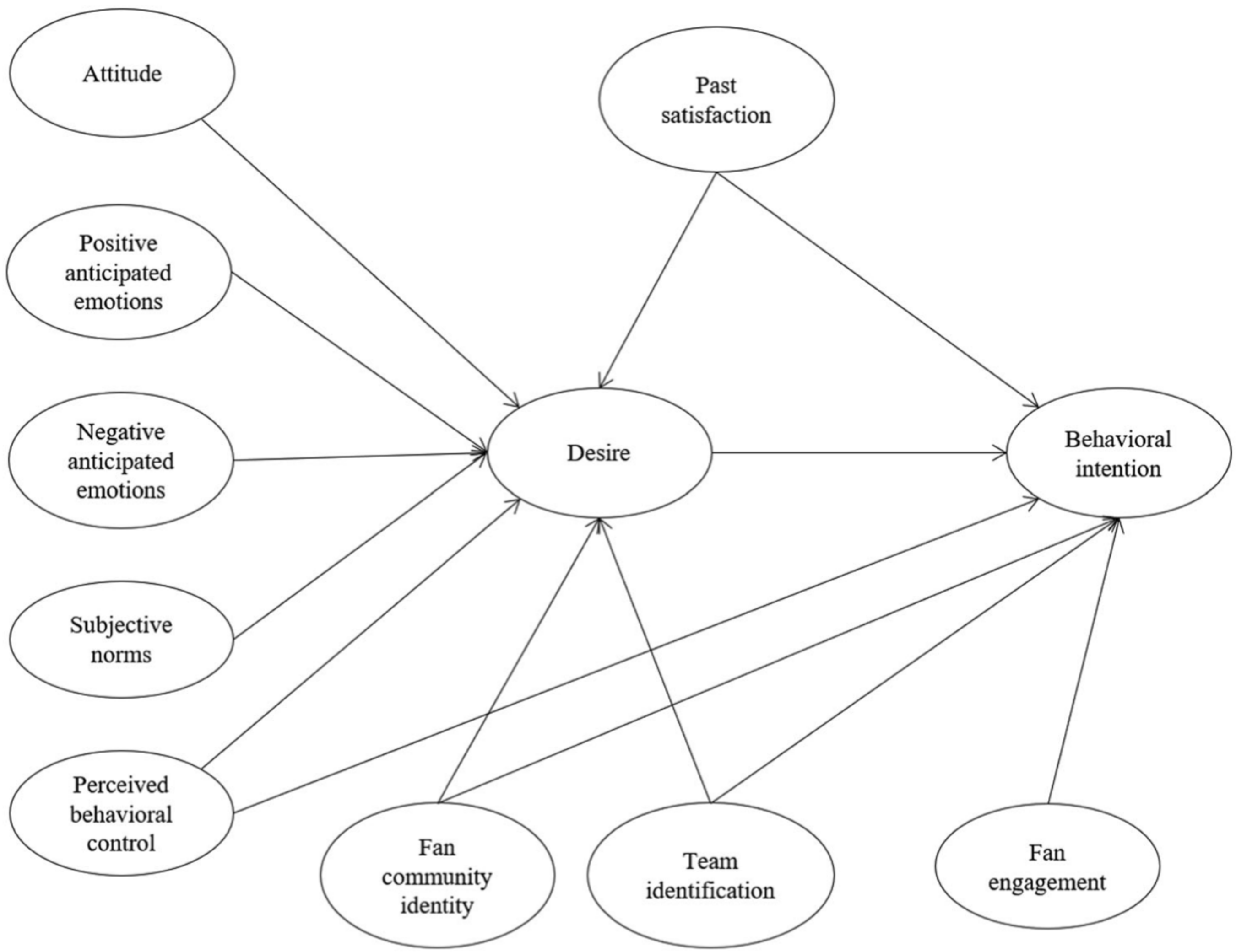


Table 1. *Descriptions of Determinants (Adapted from Yim & Byon, 2021)*

Determinant	**Description**
Attitude	The extent to which a person has a favourable or unfavourable evaluation of the behaviour.
Positive Anticipated Emotion	How the individual believes they would feel if they carried out the behaviour.
Negative Anticipated Emotion	How the individual believes they would feel if they did not carry out the behaviour.
Subjective Norms	The individuals' perception of whether peers or peer groups will approve of the behaviour.
Perceived Behavioural Control	The individuals' perception of the ease with which they can carry out the behaviour.
Fan Community Identity	The extent to which the individual feels and demonstrates a desire to belong to the team community.
Team Identification	The extent to which the individual feels psychological attached or affiliated with a particular team.
Fan Engagement	Behaviours the individual carries out to benefit their favourite team or other fans.
Past Satisfaction	The extent to which the individual had a pleasurable response to the behaviour in the past.
Desire	The motivational state that links the antecedents with intention.
Behavioural Intention	The individuals' perceived likelihood of engaging with the behaviour in the future.

Yim and Byon (2021) found that the SFMGB demonstrated strong predictive power in the contexts of event attendance, TV viewing, and social media engagement. This suggests that, whilst the model is domain-specific to sports, it is appropriately applicable to a range of contexts within this domain. Their work also compared the explanatory power of the SFMGB to that of the original MGB and the TPB across three different contexts. The variance in event attendance intentions explained by the SFMGB was 66%, whilst the MGB was slightly higher at 67% and the TPB was just 53%. In the context of TV viewing intentions, the SFMGB had the greatest explanatory power at 70%, compared with 65% for the MGB and 52% for the TPB. When it came to media consumption intentions, the SFMGB was again the strongest model, explaining 75% of the variance compared with 73% for the MGB and 34% for the TPB. These results suggest that, in most sports engagement contexts, the SFMGB explains more variance than both the MGB and the TPB, which makes it the most appropriate model for assessing sports engagement.

However, the SFMGB is also the most complicated model, and this complexity yields only marginal improvements in explanatory power. This may impact the applicability of the model and dilute the observed effects of the individual determinants due to conceptual overlaps and multicollinearity. Furthermore, this model is yet to be applied to F1, so it is unclear whether it will generalise to this unique sport. There are aspects of the SFMGB, such as team identification, that may apply differently to F1 compared with other sports. For example, the extent to which F1 fans care about the Constructors Championship has been called into question, and the Drivers' Championship is often considered to be more prestigious (e.g., Baker & Bellingham, 2024). For this reason, the concept of Team Identification in this model may be less important in F1 than in other sports. In addition, the extent to which fan engagement can be demonstrated in F1 is potentially more limited than other sports because viewing F1 in-person is a rare opportunity for fans. With only one race held in the UK each year, fans are limited in how they engage with the sport. As such, the fan engagement aspect of the SFMGB may apply differently to F1.

Nevertheless, other aspects of the SFMGB are arguably more relevant to F1 now than ever before. With the fanbase growing rapidly in recent years, the importance of community identity is evident. As highlighted by Faulkner (2024), the increased popularity has sprouted new avenues for fans to interact with one another including podcasts and social media communities. In this way, fan community identity is likely to be an important determinant for F1 engagement. The increased popularity of F1 is also likely to impact anticipated emotions. As the sport grows and certain social circles become strongly bonded through their affiliation with the sport, the so-called “fear of missing out” is likely to influence how such individuals perceive they would feel if they missed out on a race. For social fans, missing a race would not only represent a loss of enjoyment, but potentially a loss of social interaction or connection with their community which could have emotional repercussions (e.g., Liu et al., 2023). However, aside from these conceptual links, the SFMGB has never been applied to F1 before. As such, it is important that its applicability is investigated to establish the scope and validity of the model and to inform potential future research.

Data dashboards and Sports Engagement

Research linking data dashboards with sports engagement is scarce, despite being a valuable research area. As noted by Hensley (2022), a vital way of engaging populations with sports is by providing them with information and data-rich experiences. Data dashboards provide this experience with the added benefit of allowing the user freedom to explore based on their personal intrigues and interests. Cho et al. (2023) examined how different types of analytical content impacted fan engagement and found that statistics and video content significantly enhanced information acquisition and entertainment, whilst data visualisation tools facilitated social engagement with the topic. This suggests that data visualisations and statistics - a major aspect of data dashboards - can be used to enhance fan engagement. However, this study did not specifically utilise data dashboards, nor did it operationalise sports engagement. Lin et al. (2022) developed an interactive basketball-viewing prototype which embedded statistics and visualisations for the viewer, much like a data dashboard, and found that it was effective at facilitating increased understanding and engagement. This is further indication that interactive statistics and visualisations can enhance the viewing experience. Furthermore, Lo et al. (2021) examined the implementation of data visualisations at live games and found that spectators responded positively to these offerings, demonstrating increased understanding of the game without increasing mental demand. This is important because it suggests that data visualisations can be accessible, with viewers able to understand the content with ease, which is necessary if these tools are to be used to enhance engagement.

However, there is a distinct lack of research into how an interactive data dashboard may impact engagement with the sport when used non-concurrently. Despite the gap in literature, there are several theoretical ways in which a data dashboard might impact intentions to engage with F1 as defined by the SFMGB. For example, they may encourage positive attitudes towards the sport by creating accessibility and ease of understanding (e.g., Kastellec & Leoni, 2007), which may not be obtained by watching the races alone. For this reason, individuals with preconceived notions of F1 as being complicated or difficult to follow may reappraise their attitudes after accessing the data dashboard, and may also have increased PBC, thus increasing intentions to engage with F1 according to the SFMGB. Another way in which the data dashboard may help facilitate F1 engagement is through anticipated emotions. It has been suggested that a major driver of sports-viewing behaviour is past drama creating a sense of anticipation for future events (Mumford, 2012). As such, if the dashboard can highlight dramatic rivalries or accidents, this may cause the user to feel excited and curious about future races, thus increasing positive anticipated emotions and increasing intentions to engage. However, the opposite effect is also possible. If the dashboard represents predictable or uneventful outcomes, then this may cause the user to feel bored or uninterested, increasing their anticipated negative emotions and reducing their intentions to engage with the sport. Finally, data dashboard usage may also impact team identification. There is evidence that using data dashboards impacts team identification in other contexts due to increased engagement with the goals and performance of the team (e.g., Zamecnik et al., 2022). As such, the ability to analyse a teams' successes or failures in greater detail may be instrumental in facilitating increased psychological attachment and team identification. However, this effect would likely be more applicable to existing fans compared with non-fans, because individuals are unlikely to focus on a single team in significant detail unless they have an existing identification or interest in that team. As such, it is unclear how team identification would be impacted for non-fans. On one hand, they may develop an affiliation with a particular team if they admire their performance, as positive team performance has been identified as a facilitator of fandom (Wann, 2006). Conversely, they may feel alienated by the large quantity of information and avoid gravitating towards any team due to decision paralysis, a phenomenon which has been evidenced in other domains (e.g., Huber et al., 2012).

Research Questions and Hypotheses

This study seeks to address the following research questions:

1. Is the Sports-Fan Model of Goal-Directed Behaviour applicable to Formula 1?
2. Does the use of a data dashboard impact intentions to engage with Formula 1?

3. Does the impact of the data dashboard vary between fans and non-fans?

By addressing these questions, the study will contribute to the scarce research area of how models of behaviour relate to F1-engagement, and how digital tools might shape engagement, potentially informing future marketing strategies for the sport.

It is hypothesised that the SFMGB will be applicable in this context due to the discussed conceptual links between the SFMGB determinants and F1. Based on research by Cho et al. (2023), it is hypothesised that people who interact with the data dashboard will have greater intentions to engage with F1 in the future. It is expected that this impact will be present for both fans and non-fans, however its extent may vary in several ways. For example, fans may have greater intentions to engage with F1 than non-fans, regardless of whether they interacted with the data dashboard. This is because some determinants in the Sports Fan MGB such as Team Identification and Fan Community Identity represent experiences that are unique to fans of the sport, meaning non-fans are less likely to identify with these concepts. The magnitude of the impact of data dashboards on intentions may be greater for non-fans compared with fans, because increases in fan intentions may be limited by the ceiling effect (Garin, 2014). This is where fans already hold very strong viewing intentions and are limited in the extent to which their intentions can be increased further. However, the opposite is also possible. For some non-fans, using the data dashboard may only confirm their lack of interest in the sport and thus have less of a positive impact, or even a negative impact, on their intentions to engage.

Methodology

Research Approach

This study used an inductive approach, whereby the suitability of the SFMGB was assessed using the data that was collected. However, there is also scope for inductive conclusions to be drawn if the SFMGB is found to be incomplete. The study utilised quantitative methodology because intentions could be operationalised numerically using the SFMGB, allowing for statistical testing to establish conclusions. The research utilised an experimental approach with two independent groups, to which participants were allocated randomly.

Sampling

An a priori power analysis was conducted using G*Power version 3.1 for sample size estimation. With a significance criterion of $\alpha = .05$ and power = .80, the minimum sample size needed for a two-tailed independent t-test (to assess the difference in intentions between groups) was $N = 64$. The minimum sample size for a multiple linear regression with effect size = 0.15 and up to 8 determinants (i.e., the 8 determinants of desire in the SFMGB) was $N = 109$. As such, opportunity sampling was used to obtain the largest possible sample. The only inclusion criterion was that participants were over 18 years of age to ensure they could understand the content of the questionnaire and their rights as a participant. Participants were recruited through multiple means including in-person, via social media including WhatsApp and Instagram, and by a flyer that was displayed in Sheffield Student Union. Potential participants could access a Qualtrics link if they wished to take part in the study. In total, 48 participants took part in the study.

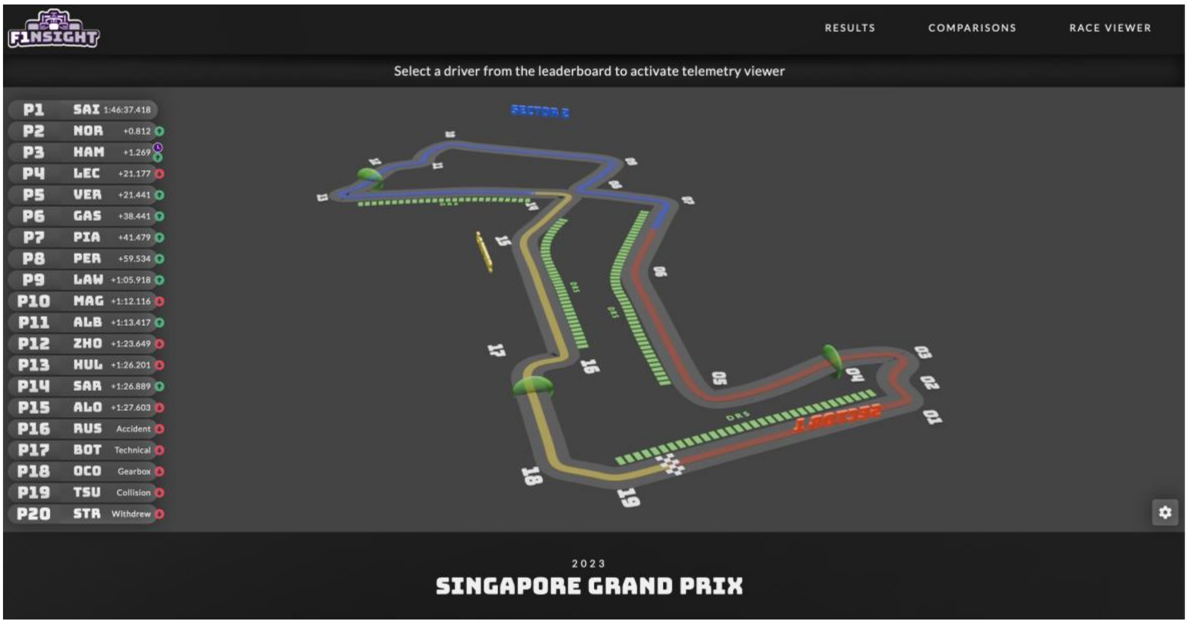
Procedure

The questionnaire was constructed using Qualtrics (https://www.qualtrics.com/) and included four blocks. The first block included the information sheet and consent form where participants could read about the study, ethical considerations and their rights as a participant, before expressing their consent or non-consent. Those who did not consent were redirected to the end of the survey, whilst those who did consent proceeded to block two. In this block, participants input some basic demographic information (Sex and Age) and expressed the extent to which they considered themselves to be fans of Formula 1. In the third block, all participants were directed to follow a link to a YouTube video on the official Formula 1 channel, which they were asked to watch in full:

Race Highlights | 2023 Singapore Grand Prix.
https://www.youtube.com/watch?v=8TmrPvoHSGQ

The video shows highlights of the 2023 Singapore Grand Prix and was necessary to contextualise the information shown in the data dashboard in the final block. The 2023 season was chosen because it is recent enough to be relevant and representative of the current sport, but not too recent to be remembered in detail by fans. The Singapore Grand Prix was chosen because it contains multiple incidents and points of analysis for viewers to consider, without being unrepresentative of a typical race. For the final block, participants were split randomly into either the experimental or control group. The experimental group was directed to follow a link to a data dashboard about the 2023 Singapore Grand Prix (https://www.f1nsight.com/#/race/1219), as shown in Figure 2.

Figure 2. *Data Dashboard by F1NSIGHT*



This is a dashboard by F1NSIGHT, chosen because it has a range of functionalities and can be accessed online without the need for any additional software. Users can scroll down on the dashboard to access information such as position data, tyre strategies, and pit stops. Participants were asked to freely interact with the dashboard for as long as they wished, as this is most representative of how people may use a data dashboard in their own lives. Participants then proceeded to complete the SFMGB questionnaire. Those in the control group were not directed to the data dashboard and instead completed the SFMGB questionnaire immediately after watching the video. This allowed for comparison of responses between the experimental and control groups, to determine the impact of access to the data dashboard. The SFMGB questionnaire itself was created based on recommendations by Yim and Byon (2021), containing 31 Likert-style questions as shown in Appendix A.

Measures

SFMGB variables were computed by summing the values of each respective questionnaire item. For items that were reverse scored, the values were calculated using the following formula:

$$Value \ = (Min + Max) - \sum items$$

Where *Value* is the computed variable, *Min* is the minimum value of the sum of items and *Max* is the maximum value of the sum of items.

For example, the SN variable was computed by summing the two reverse-scored Subjective Norms items, and then subtracting from 16, such that the lowest possible value of 2 became the highest possible value of 14, and vice-versa. This produced a score out of 14 for this construct.

An additional variable “Group” was computed to represent whether the respondent was part of the dashboard or non-dashboard group, where 0 represented the non-dashboard group and 1 represented the dashboard group.

Furthermore, a “FanStatus” variable was computed such that those who “Definitely” or “Probably” considered themselves F1 fans were given a FanStatus value of 1, whilst all other participants were considered non-fans and given a FanStatus value of 0.

A summary of each SFMGB construct measured by the questionnaire, along with their corresponding items and value range is shown in Table 2.

Table 2. *SFMGB Constructs Summary*

Construct	Description of Items	Adapted from	Notes	Items in Appendix A	Computed Value Range
Attitudes	Three items on 7-point scales.	Osgood et al. (1957)		1, 2, 3	3 - 21
Subjective Norms	Two items on 7-point scales.	Perugini and Bagozzi (2001)	Reverse Scored	4, 5	2 - 14
Perceived Behavioural Control	Two items on 7-point scales.	Bagozzi et al. (2006)		6, 7	2 - 14
Positive Anticipated Emotions	Three items on 7-point scales.	Perugini and Bagozzi (2001)	Reverse Scored	8, 9, 10	3 - 21
Negative Anticipated Emotions	Four items on 7-point scales.	Perugini and Bagozzi (2001)	Reverse Scored	11, 12, 13, 14	4 - 28
Desire	Three items. Two using 11-point scales and one using a 6-point scale.	Perugini and Bagozzi (2001)		15, 16, 17	3 - 28
Fan Engagement	Four items on 7-point scales.	Yoshida et al. (2014)	Some items used by Yoshida et al. were not applicable, so an appropriate reduced set was used.	18, 19, 20, 21	4 - 28

Construct	**Description of Items**	**Adapted from**	**Notes**	**Items in Appendix A**	**Computed Value Range**
Fan Community Identity	Two items on 5-point scales.	Bergami & Bagozzi (2000); Bagozzi et al. (2006)		22, 23	2 - 10
Team Identification	Three items on 7-point scales.	Trail and James (2001)		24, 25, 26	3 - 21
Past Satisfaction	Two items on 7-point scales.	Yoshida and James (2010)		27, 28	2 - 14
Behavioural Intentions	Three items on 7-point scales.	Yim and Byon (2021)		29, 30, 31	3 - 21

Ethical Considerations

This study collected primary data from participants using a questionnaire. Since the topic in question was not culturally or politically sensitive the risk level was low. For participants to give informed consent, there was an information sheet prior to the questionnaire which explained what participants would be asked to do. Any participants who were comfortable with taking part after reading this information could give their consent and proceed with the questionnaire. For the duration of the questionnaire, participants had the right to withdraw. Participants were made aware in the introduction that they could close their window at any time before completing the questionnaire and their data would not be submitted. However, they were also made aware that once their response had been submitted it would not be possible for them to withdraw as their data would be immediately anonymised. All data was kept anonymous and confidential. The research was ethically reviewed and received approval from the Information School at The University of Sheffield.

Data Analysis

At the end of the collection period, data was exported to SPSS for analysis and a series of statistical tests were undertaken.

Firstly, attrition analyses were conducted to assess whether there were any significant demographic differences between those who engaged fully with the survey and those who did not. Next, descriptive statistics were calculated to establish the key metrics of all variables of interest. To address the first research question ("Is the Sports-Fan Model of Goal-Directed Behaviour applicable to Formula 1?"), correlation, mediation, and multiple regression analyses were conducted to assess the applicability of the SFMGB. To address the second research question ("Does the use of a data dashboard impact intentions to engage with

Formula 1?"), an independent t-test was conducted to compare the measure of intentions between the control and experimental groups. For the final research question ("Does the impact of the data dashboard vary between fans and non-fans?"), a Two-Way ANCOVA was conducted to assess whether the effect of the data dashboard on intentions differed depending on fan status.

Table 3 shows the abbreviated names of the SFMGB variables as they appear in the analyses.

Table 3. *Variable Names*

Variable Name	Corresponding SFMGB Construct
Att	Attitudes
SN	Subjective Norms
PBC	Perceived Behavioural Control
PAE	Positive Anticipated Emotions
NAE	Negative Anticipated Emotions
Desire	Desire
FanEng	Fan Engagement
ComId	Fan Community Identity
TeamId	Team Identification
PastSat	Past Satisfaction
Int	Behavioural Intention

Results

Participants

A total of 48 participants took part in the study. However, 12 responses (25%) were excluded because of non-completion of the SFMGB questionnaire. This resulted in 36 valid responses for analyses of the SFMGB.

Additional screening showed that a further 14 participants (38.9%) did not engage meaningfully with the video, with recorded timings suggesting they watched the video for less than 30 seconds. As the video was essential for contextualising the data dashboard, these participants were excluded from dashboard-related analyses. 30 seconds was chosen as a minimum engagement time as it represents a level of genuine engagement which ensures valid contextual understanding of the dashboard. This left 22 participants for analyses examining the impacts of the data dashboard: 14 in the experimental group (who viewed the dashboard), and 8 in the control group (who did not).

Of the 36 participants included in the primary analyses, 18 were Male (50%), 17 were Female (47%) and 1 was non-binary or a third gender (2.8%). The age of participants ranged from 19 to 61 (M = 25.3, SD = 9.86).

In terms of self-reported fandom, 17 participants reported as being “Definitely” fans of F1 (47.2%), 5 reported as “Probably” (13.9%), 4 as “May or may not” (11.1%), 6 as “Probably Not” (16.7%), and 4 as “Definitely Not” (11.1%). Based on these responses, 22 participants (61.1%) were categorised as fans, and 14 (38.9%) as non-fans for group comparison purposes.

Attrition Analyses

To assess potential bias due to attrition, comparisons were made between participants who sufficiently engaged with the video and those who did not.

An independent samples t-test revealed no significant difference in age between those who engaged (M = 25.82, SD = 9.79) and those who did not (M = 24.43, SD = 10.29; $t(34)$ = -0.407, p = .686). A chi-square test of independence showed no significant association between gender and engagement ($X^2(2, N = 36)$ = .963, p = .618). Similarly, there were no significant differences in fandom between the two groups ($X^2(4, N = 36)$ = 2.890, p = .576).

These results suggest that attrition was not systematically associated with demographic or fandom-related characteristics.

Descriptive Statistics

Descriptive statistics for the key SFMGB variables are presented in Table 4. Scores reflect participants' responses on each construct as measured by the SFMGB.

Table 4. *Descriptive Statistics for SFMGB (N = 36)*

	M	*SD*
Attitudes	15.92	3.91
Subjective Norms	10.92	2.31
Perceived Behavioural Control	11.72	2.09
Positive Anticipated Emotion	15.72	4.26
Negative Anticipated Emotion	10.69	6.87
Desire	18.86	8.78
Fan Engagement	15.36	7.85
Fan Community Identity	4.94	2.50
Team Identification	11.69	6.36
Past Satisfaction	10.58	2.73
Intentions	14.83	5.99

Correlating SFMGB Constructs

Table 5 presents the Pearson correlation coefficients among all constructs in the SFMGB.

Table 5. *Correlation Matrix*

	1	2	3	4	5	6	7	8	9	10
1. Att	-									
2. SN	.626**	-								
3. PBC	.315	.308	-							
4. PAE	.778**	.384*	.225	-						
5. NAE	.479**	.319	.370*	.592**	-					
6. Desire	.798**	.388*	.197	.887**	.626**	-				
7. FanEng	.758**	.479**	.394*	.800**	.685**	.853**	-			
8. ComId	.647**	.480**	.484**	.654**	.678**	.720**	.907**	-		
9. TeamId	.848**	.615**	.487**	.786**	.523**	.816**	.865**	.779**	-	
10. PastSat	.800**	.466**	.479**	.761**	.414*	.741**	.754**	.672**	.751**	-
11. Intentions	.786**	.348*	.316	.829**	.636**	.872**	.908**	.782**	.811**	.814**

Note. * $p < .05$., ** $p < .01$. (2-Tailed)

As shown, most variables are significantly and positively correlated with one another, particularly Attitudes, PAE, Desire, Fan Engagement, and Intentions, which all exhibit strong intercorrelations ($r > .70$, $p < .01$). However, these strong correlations raise concerns regarding multicollinearity, which may affect the interpretability of coefficients in regression and mediation models. Variance Inflation Factors (VIFs) will be warranted in subsequent analyses to assess the extent of multicollinearity

Mediation Analyses

To assess the validity of the SFMGB in the context of F1, a series of mediation analyses were conducted using Model 4 of the PROCESS macro in SPSS (Model 4; Hayes, 2022). In each analysis, Desire was specified as the mediator and Intention as the Dependent Variable, with each determinant entered individually as the Independent Variable. Analyses were first conducted without covariates to assess basic mediation effects, and then they were repeated with all remaining determinants as covariates to assess the unique contribution of each IV to Desire and Intention.

The full mediation model explained a significant 91.5% of the variance in intention (R^2 = .915), and the overall model predicting Desire was also significant (R^2 = .887, $F(9, 26)$ = 22.68, $p < .001$).

Independent Variable: Attitude

The simple mediation analysis revealed a significant indirect effect (b = .823, BootCI = [.398, 1.23]), suggesting that Desire mediates the relationship between Attitude and Intention when other determinants are not controlled.

However, in the adjusted model with all other determinants entered as covariates, it was found that Attitude did not significantly predict Desire (b = .245, $t(26)$ = .66, p = .512).
In the combined regression model predicting Intention (with both Attitude and Desire as predictors and all covariates included), the model explained a significant amount of variance in Intention, R^2 = .915, $F(10, 25)$ = 26.76, $p < .001$. However, neither Attitude (B = .183, $t(25)$ = .82, p = .423) nor Desire (B = .022, $t(25)$ = .18, p = .856) had a significant direct effect on Intention.

Bootstrapping with 5,000 samples revealed **no significant indirect effect** of Attitude on Intention through Desire (b = .005, BootCI = [-.108, .165]) indicating that Desire **does not mediate** the relationship between Attitude and Intention in the presence of other covariate determinants.

Independent Variable = SN

The simple mediation analysis with no covariates revealed a significant indirect effect (b = .874, BootCI = [.040, 1.87]), suggesting that Desire mediates the relationship between SN and Intention when all other determinants are not controlled.

However, it was found that SN did not significantly predict Desire after controlling for the covariates (B = -.540, $t(26)$ = -1.53, p = .138). In the combined regression model predicting Intention (with both SN and Desire as predictors and all covariates included), SN had a significant direct effect on Intention (B = -.579, t(25) = -2.76, p = .011). This suggests that SN has a direct negative impact on intentions when controlling for the other determinants.

Bootstrapping with 5,000 samples revealed **no significant indirect effect** of SN on Intention through Desire (b = -.012, BootCI = [-.355, .147]) indicating that Desire **does not mediate** the relationship between Subjective Norms and Intention in the presence of other covariate determinants.

Independent Variable = PBC

The simple mediation analysis with no covariates revealed no significant indirect effect (b = 0.475, BootCI = [-.230, 1.16]), suggesting that Desire does not mediate the relationship between PBC and Intention.

Nevertheless, it was found that PBC significantly predicted Desire after controlling for the covariates (B = -1.01, t(26) = -2.33, p = .028). In the combined regression model predicting Intention (with both PBC and Desire as predictors and all covariates included), PBC did not have a significant direct effect on Intention (B = -.388, t(25) = -1.345, p = .191).

Bootstrapping with 5,000 samples revealed **no significant indirect effect** of PBC on Intention through Desire (b = -.022, BootCI = [-.489, .277]) indicating that Desire **does not mediate** the relationship between PBC and Intention in the presence of other covariate determinants.

Independent Variable = PAE

The simple mediation analysis with no covariates revealed a significant indirect effect (b = .798, BootCI = [.299, 1.23]), suggesting that Desire mediates the relationship between PAE and Intention.

However, after including covariates in the mediation it was found that PAE did not significantly predict Desire (B = .54, t(26) = 1.69, p = .102). In the combined regression model predicting Intention (with both PAE and Desire as predictors and all covariates included), PAE did not have a significant direct effect on Intention (B = -.093, t(25) = -.45, p

= .654). Bootstrapping with 5,000 samples revealed **no significant indirect effect** of PAE on Intention through Desire (b = .012, BootCI = [-.279, .257]). This indicates that Desire **does not mediate** the relationship between Positive Anticipated Emotions and Intention in the presence of other covariates.

Independent Variable = NAE

The simple mediation analysis with no covariates revealed a significant indirect effect (b = .425, BootCI = [.218, .716]), suggesting that Desire mediates the relationship between NAE and Intention.

However, after including covariates, It was found that NAE did not significantly predict Desire (B = .241, t(26) = 1.67, p = .107). In the combined regression model predicting Intention (with both NAE and Desire as predictors and all covariates included), NAE did not have a significant direct effect on Intention (B = .142, t(25) = 1.55, p = .133). Bootstrapping with 5,000 samples revealed **no significant indirect effect** of NAE on Intention through Desire (b = .005, BootCI = [-.047, .169]), indicating that Desire **does not mediate** the relationship between Negative Anticipated Emotions and Intention when other determinants are controlled.

Independent Variable = FanEng

The simple mediation analysis with no covariates revealed a significant indirect effect (b = .232, BootCI = [.043, .391]), suggesting that Desire mediates the relationship between Fan Engagement and Intention.

However, after controlling for covariates, it was found that Fan Engagement did not significantly predict Desire (B = .166, t(26) = .61, p = .547). Nevertheless, in the combined regression model predicting Intention (with both Fan Engagement and Desire as predictors and all covariates included), Fan Engagement had a significant direct effect on Intention (B = .423, t(25) = 2.55, p = .017). Bootstrapping with 5,000 samples revealed **no significant indirect effect** of Fan Engagement on Intention through Desire (b = .004, BootCI = [-.175, .102]), indicating that Desire **does not mediate** the relationship between Fan Engagement and Intention when other determinants are controlled.

Independent Variable = ComId

The simple mediation analysis with no covariates revealed a significant indirect effect (b = 1.109, BootCI = [.657, 1.603]), suggesting that Desire mediates the relationship between Fan Community Identity and Intention.

However, when including covariates, it was found that ComId did not significantly predict Desire (B = .001, t(26) = .00, p = .999). In the combined regression model predicting Intention (with both ComId and Desire as predictors and all covariates included), ComId also did not have a significant direct effect on Intention (B = –.230, t(25) = –.61, p = .545). Bootstrapping with 5,000 samples revealed **no significant indirect effect** of ComId on Intention through Desire (b = .000, BootCI = [–.254, .299]), indicating that Desire **does not mediate** the relationship between Fan Community Identity and Intention when other determinants are controlled.

Independent Variable = TeamId

The simple mediation analysis with no covariates revealed a significant indirect effect ($b = .483$, BootCI = [.205, .825]), suggesting that Desire mediates the relationship between Team Identification and Intention.

However, it was found that TeamId did not significantly predict Desire after controlling for covariates ($B = .506$, $t(26) = 1.75$, $p = .092$). In the combined regression model predicting Intention (with both TeamId and Desire as predictors and all covariates included), TeamId also did not have a significant direct effect on Intention ($B = .167$, $t(25) = .90$, $p = .377$). Bootstrapping with 5,000 samples revealed **no significant indirect effect** of TeamId on Intention through Desire ($b = .011$, BootCI = [–.138, .337]), indicating that Desire **does not mediate** the relationship between Team Identification and Intention when other determinants are controlled.

Independent Variable = PastSat

The simple mediation analysis with no covariates revealed a significant indirect effect ($b = .970$, BootCI = [.557, 1.469]), suggesting that Desire mediates the relationship between Past Satisfaction and Intention.

However, it was found that PastSat did not significantly predict Desire after controlling for covariates ($B = .540$, $t(26) = 1.09$, $p = .284$). In the combined regression model predicting Intention (with both PastSat and Desire as predictors and all covariates included), Past Satisfaction had a significant direct effect on Intention ($B = .785$, $t(25) = 2.57$, $p = .017$). Bootstrapping with 5,000 samples revealed **no significant indirect effect** of PastSat on Intention through Desire ($b = .012$, BootCI = [–.196, .379]), indicating that Desire **does not mediate** the relationship between Past Satisfaction and Intention when other determinants are controlled.

Does Desire predict Intention?

To determine whether Desire significantly predicts Intention as suggested by the SFMGB, a multiple regression analysis was conducted with Intention as the Dependent Variable. The predictors included and Desire, PBC, Fan Community Identity, Team Identification, Fan Engagement and Past Satisfaction.

Desire did not emerge as a significant predictor of Intention (B = .156, p = .132), suggesting it may not exert a unique influence on intention when other relevant variables are considered.

Table 6. *Model Summary*

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.942[a]	.888	.865	2.20174

a. Predictors: (Constant), PastSat, PBC, ComId, TeamId, Desire, FanEng

Table 7. *Multiple regression on intention*

Variable	B	p	Collinearity (VIF)
Desire	0.156	0.132	5.637
PBC	-0.122	0.633	2.027
ComId	-0.290	0.456	6.605
TeamId	-0.055	0.690	5.479
FanEng	0.522	0.004*	12.415
PastSat	0.603	0.018*	3.097

*$p < .05$

It is noteworthy that the VIF scores exceeded 2 for all predictors, with some values above 5 and one exceeding 10. This indicates a potentially high degree of multicollinearity, which may obscure the unique contributions of individual predictors by inflating standard errors and suppressing significant effects.

Overall, the multiple regression model explained 88.9% of the variance in intention (R^2 = .888).

Summary

Table 8 summarises the significance of every path of the SFMGB, as labelled in Figure 3. The only significant paths were found to be from PBC to Desire, Fan Engagement to Intention, and Past Satisfaction to Intention.

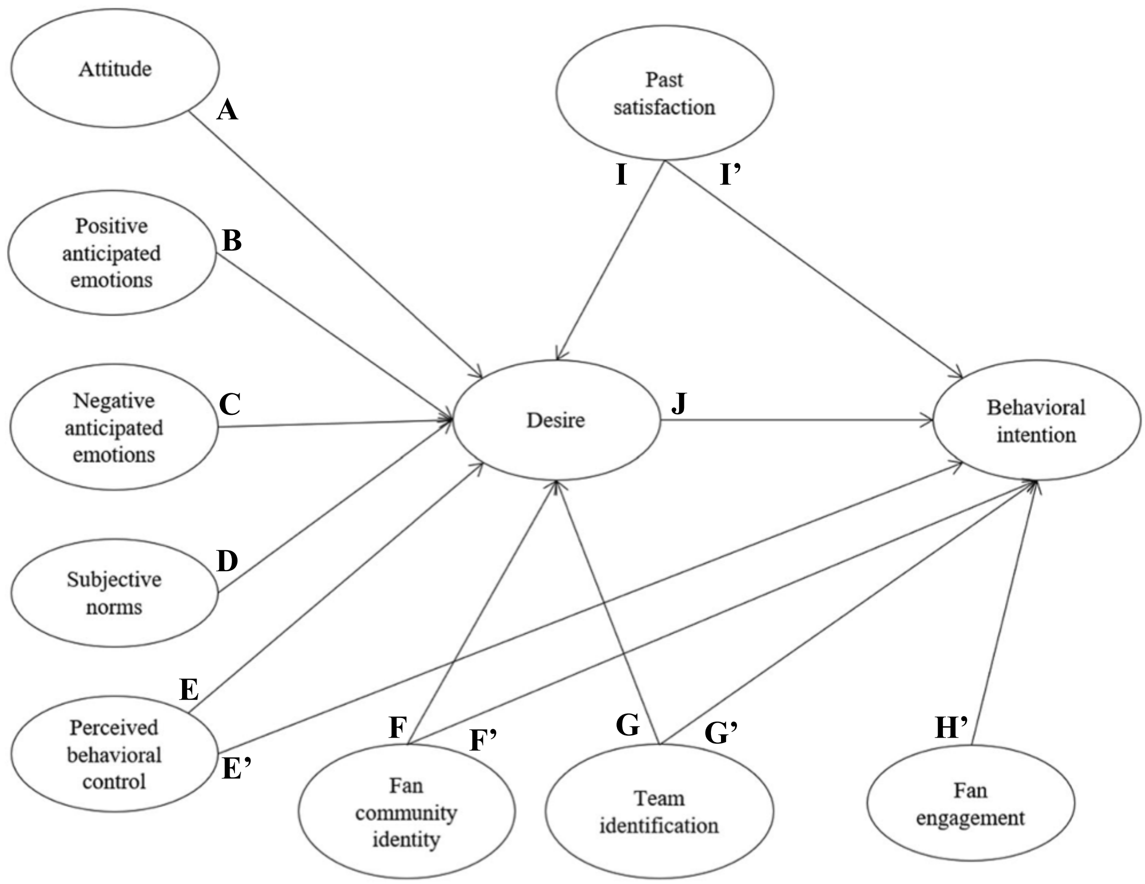
Table 8. *Summary of Paths*

Path	**B**	**SE**	**t**	**p**
A: Attitude >> Desire	0.245	0.368	0.66	.512
B: PAE >> Desire	0.544	0.321	1.69	0.102
C: NAE >> Desire	0.241	0.144	1.67	0.107
D: SN >> Desire	-0.540	0.353	-1.53	0.138
E: PBC >> Desire	-1.009	0.433	-2.33	0.028*
E’: PBC >> Intention	-0.388	0.288	-1.35	0.191
F: FCI >> Desire	0.009	0.619	0.002	0.999
F’: FCI >> Intention	-0.230	0.375	-0.614	0.545
G: Team Id >> Desire	0.506	0.289	1.75	0.092
G’: Team Id >> Intention	0.167	0.185	0.900	0.377
H’: Fan Eng >> Intention	0.423	0.166	2.55	0.017*
I: Past Sat >> Desire	0.540	0.494	1.09	0.284
I’: Past Sat >> Intention	0.785	0.306	2.57	0.017*

Path	B	SE	t	p
J: Desire >> Intention	0.156	0.101	1.55	0.132

*p < .05

Figure 3. *SFMGB Paths*



Independent Samples T-Test

An independent samples t-test was conducted to determine whether there was any significant difference in the F1 engagement intentions of those in the dashboard group compared with those in the non-dashboard group.

Table 9. *Group Statistics*

	Dashboard Status	N	Mean	Std. Deviation	Std. Error Mean
Intentions	No Dashboard	8	12.38	7.29	2.58
	Dashboard	14	14.29	5.94	1.59

Table 10. *Independent Samples T-Test*

	t	*df*	*p*	Mean Difference	Std. Error Difference	95% CI	
						Lower	Upper
Intentions	-.669	20	.256	-1.91	2.86	-7.87	4.05

There was no significant difference in intentions between the dashboard group (M = 14.29, SD = 5.94) and the non-dashboard control group (M = 12.38, SD = 7.29; t(20) = -0.669, p = 0.256), despite the dashboard group demonstrating slightly higher mean intentions. This suggests that participants' exposure to the data dashboard did not meaningfully impact their intentions to engage with F1.

Two-Way ANCOVA

To examine whether the effect of the data dashboard on intentions differed depending on fan status, a two-way ANCOVA was conducted. The model included Group (dashboard vs no dashboard) and Fan Status (fan vs non-fan) as independent variables, with Intention as the dependent variable.

Table 11. *Tests of Between-Subjects Effects*

Source	Type III Sum of Squares	df	Mean Square	F	Sig.
Corrected Model	595.547[a]	3	198.516	14.081	<.001
Intercept	3048.838	1	3048.838	216.254	<.001
Group	62.792	1	62.792	4.454	.049
FanStatus	574.167	1	574.167	40.726	<.001
Group * FanStatus	31.935	1	31.935	2.265	.150
Error	253.771	18	14.098		
Total	4913.000	22			
Corrected Total	849.318	21			

Dependent Variable: Int

a. R Squared = .701 (Adjusted R Squared = .651)

The results showed a significant main effect of Group, $F(1, 18) = 4.45$, $p = .049$, suggesting that overall, exposure to the data dashboard has a positive impact on engagement intentions when controlling for fan status.

There was also a strong main effect of Fan Status, $F(1, 18) = 40.73$, $p < .001$. This indicates that self-identified fans of F1 reported significantly higher engagement intentions than non-fans.

However, the interaction between Group and Fan Status was not significant, $F(1, 18) = 2.27$, $p = .150$. This suggests that the effect of dashboard exposure on intentions did not differ significantly between fans and non-fans. Overall, the model explained a substantial 70.1% of the variance in Intention ($R^2 = .701$, Adjusted $R^2 = .651$), indicating that dashboard exposure and fan status are meaningful predictors of intention.

Discussion

Aims and Findings

Q1: Does the SFMGB apply to Formula 1?

The SFMGB builds upon the MGB by adding additional determinants such as Fan Community Identity and Team Identification. These determinants were hypothesised to have a unique impact in the context of Formula 1 because it is a sport which has fostered a high degree of inclusivity (Hamlin, 2022), whilst also being predominantly individual-focused rather than team-focused (Baker & Bellingham, 2024). However, this study found limited support for the applicability and validity of the SFMGB within the context of Formula 1. Among the hypothesised paths, only three were statistically significant: from PBC to Desire, from Fan Engagement to Intention, and from Past Satisfaction to Intention. While the model as a whole predicted Desire, most individual determinants did not demonstrate statistically significant predictive power. This suggests that there may be a high degree of multicollinearity, where the effects of some determinants are diluted by the effects of other conceptually similar determinants. There is extensive evidence that multicollinearity can have detrimental effects on the interpretation of both regression (e.g., Mundfrom et al., 2018; Adeboye et al., 2014) and mediation (e.g., Yoo et al., 2014; Iacobucci, 2008) analyses, meaning the present findings may have been compromised.

Notably, the overall model accounted for 88.9% of the variance in Intention, a figure that exceeds the explanatory power reported in prior research by Yim and Byon (2020) who reported R^2 values between 66% and 75%. This substantial R^2 value may be a sign of model overfitting, particularly given the small sample size utilised in the study (N = 36) in combination with the high quantity of predictors included in the SFMGB. These factors can increase the risk that the model captures random variation in the sample as opposed to robust effects (e.g., Babyak, 2024).

These findings raise questions about the practical utility of the SFMGB in the context of Formula 1. Yim and Byon (2021) found the SFMGB to demonstrate only marginal improvements in predictive power over the standard MGB, and the present study suggests that the substantial increase in the quantity of determinants may make it more difficult to identify individual effects. Since Fan Engagement was the only additional determinant in the SFMGB which significantly predicted intentions, the inclusion of the other additional determinants may not substantially enhance the model's explanatory power compared to the standard MGB. Future research could utilise larger samples and explore dimension reduction techniques like factor analysis or stepwise regression to isolate the most impactful predictors. This may clarify which components of the SFMGB are essential and which may be redundant in the context of motorsport fan behaviour.

Q2: Does the use of a data dashboard impact intentions to engage with Formula 1?

The mean intention score of participants who were shown the dashboard was slightly greater than the mean intention score of the control group, but initial results from an independent samples t-test found this difference to be non-significant. Nevertheless, a subsequent two-way ANCOVA which controlled for fan status found that those in the dashboard group did have significantly greater intentions to engage compared with the non-dashboard group. This discrepancy between the t-test and ANCOVA suggests that fan status may act as a confounding variable, masking the true effect of the dashboard when not accounted for. In controlling for this variable, the ANCOVA provides stronger evidence that dashboard usage independently contributes to increased behavioural intention.

As such, it was found that the use of a data dashboard did modestly impact intentions to engage with Formula 1. Although this was the first study to address data dashboards and engagement in Formula 1 specifically, there is research in parallel domains that could corroborate this finding. For example, data dashboards have been demonstrated to be effective in increasing the accessibility and transparency of data in non-sport contexts, and thus improving engagement with the content (e.g., Schwendimann et al., 2016; Rabiei & Almasi, 2022). Furthermore, Cho et al. (2023) investigated how data visualisations could impact engagement with sports and discovered increased fan engagement when the sport offered viewers data visualisations. Although this study did not include Formula 1 visualisations, it nevertheless supports the use of data visualisation tools in sports more generally. Overall, this finding supports suggestions by Hensley (2022) that sports fans should be provided more analytical content to foster engagement. This signals the potential value of utilising a data visualisation dashboard to expand F1's fan base and viewer engagement.

Q3: Does the impact of the data dashboard vary between fans and non-fans?

A two-way ANCOVA revealed no significant interaction between Group (dashboard vs no-dashboard) and Fan Status (fan vs non-fan), indicating that the effect of the data dashboard on intentions did not differ significantly based on whether participants were fans. This finding suggests that the positive influence of data visualisation dashboards on F1 engagement may be equally applicable to both casual and enthusiastic viewers and thus may be an effective strategy in both attracting new viewers and ensuring fan retention. This means the ceiling effect (Garin, 2014), which was hypothesised as a possible mechanism by which the impact of the data dashboard may be limited for existing fans compared with non-fans, was not necessarily applicable in this context. Instead, it suggests that existing fans still provide an opportunity to increase engagement intentions. This finding also suggests that non-fans were not discouraged from engaging with Formula 1 through the increased detail provided to them by the dashboard. This challenges previous research by Zheng & Chen (2022) who found that non-fans do not gain additional enjoyment from data visualisations due to the complexity of the information, which increases their cognitive load without meaningful benefits. Nevertheless, it is worth considering that non-fans are unlikely to engage with dashboards naturally. In the present study, all participants were encouraged to engage, and this was made easy by providing a direct link to the dashboard. In natural contexts, however, they would have to seek out this information deliberately. This is an action that fans are more likely to make than non-fans (Cho et al., 2023; Lin et al., 2022; Daraojimba et al., 2024). As such, whilst data dashboards may be effective at fostering engagement regardless of fandom, fans are more likely to actively seek out such tools than non-fans, which impacts the true implications of these findings.

Limitations and Future Directions

The results of this study should be interpreted in the context of its methodological limitations. Firstly, sampling limitations meant that only 36 participants were included in any analyses, which could significantly impact the validity of the results. It has been suggested by Ioannidis (2008) that many associations discovered in early studies are over-inflated due to underpowering. That is, the study does not have a large enough sample size to reliably detect an effect. Indeed, an A-Priori analysis revealed that the minimum sample size required to reach 80% power in any analyses in the present study was 64. An underpowered study risks Type-I errors (false positives) by causing increased variability in effect size estimates, increasing the likelihood that an effect appears more extreme than it really is. It also risks Type-II errors (false negatives) because, even if an effect exists in the population, the sample size may be too small to be representative of the population, thus reducing the likelihood of an effect being found. The implication for the present study is that the lack of evidence for mediation in the SFMGB could be a result of Type-II errors, rather than a true indication of the model's inapplicability to F1. The sample size was particularly small for the ANCOVA (N = 22) which increases the risk that the impact of the data dashboard was the result of a Type-I error. Further research with a larger and more diverse sample is recommended to validate these findings.

The present study is also limited in how the stimuli were delivered. It was assumed that participants paid attention to the F1 video and engaged with the dashboard in a natural way, but this may not be the case. To ensure that only genuine participants were included in analyses, those who watched the video for less than 30 seconds were removed. Research suggests that there is no specific length of time after which people disengage with uninteresting content. Instead, it is highly dependent on the individual and the context (e.g., Westgate, 2019). As a result, the chosen 30 second minimum engagement time was largely arbitrary – a time chosen to identify the obvious non-engagers whilst retaining a modest sample size. Overall, participants engaged with the video for an average of 158 seconds. After filtering out the non-engagers, the mean engagement time increased to 251 seconds. Whilst this is a substantial improvement, the video itself was approximately 480 seconds in length, meaning these participants only engaged with 52% of the video on average. There was not a minimum engagement time for dashboard usage because this would reduce the ecological validity of the methodology; in real life, participants would be free to use the dashboard for as much or as little time as they wished. The lack of video engagement is a limitation because the information presented by the dashboard relied on the events shown in the video, without which the participants would not be able to sufficiently understand the dashboard content. More robust methods should be used in future to ensure video engagement. These could include comprehension questions to ensure the participant has sufficiently understood the video, or eye-tracking or observation techniques to assess whether the participant has paid attention. Whilst comprehension questions could be implemented without affecting the overall design of the study, observation techniques would require a more controlled laboratory setting, and the research method of future research should be considered with these adaptations in mind.

Demand characteristics may have also confounded the effects of the dashboard; participants who were exposed to both the video and dashboard may have subconsciously responded more positively, due only to the quantity of stimuli rather than the dashboard itself. This idea is supported by research, although the exact effect this has on responses is highly context-dependent (McCambridge et al., 2012). This is a limitation because it is unclear whether any observed effect was truly a result of the data dashboard. To control these demand characteristics, future research may benefit from a different experimental design. For example, the control group could be given an unrelated dashboard rather than no dashboard. This normalises the quantity of stimuli across both groups, eliminating the possibility that the quantity of stimuli impacts responses.

The length of the questionnaire also presents a potential issue, especially considering that no compensation could be offered in this study. The questionnaire was estimated to take up to 20 minutes to complete, which may impact the quality of responses compared with shorter questionnaires. Galesic and Bosnjak (2009) examined the quality of web survey responses across 10-, 20- and 30-minute timeframes and found that longer surveys led to shorter and more uniform responses, suggesting that participants start to lose interest when faced with excessively long surveys. There is limited evidence that compensation improves data quality, yet there is research to suggest that it improves response rates (More et al., 2022) which would positively impact the sample size. Whilst it was not possible to reduce the length of the SFMGB questionnaire without impacting its validity, responses may have been improved if the video and dashboard sections of the questionnaire were shorter. The lack of engagement with the video suggests that participants may have already become bored at this early stage of the questionnaire, which may have impacted their responses. Alternatively, providing compensation could have drastically improved completion rates and thus improved statistical power. Future research should consider these aspects to ensure veracity and volume of the data.

A major limitation of models like the SFMGB, which have their roots in the Reasoned Action Approach (Fishbein & Ajzen, 2010), is that they assume intentions to be a reliable predictor of behaviour. However, there is evidence to suggest that intentions do not equate to behaviour, as variation in intentions only explains around 25% of the variation in behaviour (e.g., Webb & Sheeran, 2006; Sheppard, Hartwick & Warshaw, 1988). The difference between intentions and behaviour is known as the intention-behaviour gap (I-B gap) and suggests that there is a significant proportion of behaviour which cannot be attributed to intentions. This undermines psychological models which predict behaviour from intentions. For this reason, it is unlikely that the increase in F1-engagement intentions for the dashboard group would translate equivalently into real-life engagement behaviour. There are also mechanisms at play which may increase the I-B gap in the present context, particularly for non-fans. There is evidence that habits impact the relationship between intentions and behaviour, such that the I-B gap is greater for non-habitual behaviours (Gardner et al., 2020). It could be argued that F1 fans engage with the sport habitually, and thus their engagement intentions could be more likely to translate into behaviour compared with non-fans. Indeed, the MGB addressed this by including past behaviour as a direct predictor of behaviour. As such, the true implications of the findings relating to engagement intentions are unclear and may differ by demographic. A further mechanism by which the I-B gap may be greater for non-fans is through their PBC. The MGB includes PBC as a predictor of both intentions and behaviour because it is assumed to be somewhat representative of the actual control a person has over the behaviour. However, it is possible that non-fans have a reduced understanding of their behavioural control due to their lack of familiarity with the sport. In the UK, Sky Sports is the exclusive rights-holder for Formula 1 meaning fans must pay a significant subscription fee to watch the sport. Whilst F1 fans are inevitably familiar with this system, non-fans may be unaware that the sport cannot be accessed through other channels. As such, non-fans may have overestimated their behavioural control as they could not anticipate this as an obstacle in their engagement. The result is that, when it comes to translating intentions into behaviour, non-fans may face added difficulties which reduce their likelihood of engaging with the sport, thus increasing their I-B gap. Nevertheless, research by Mumford (2012) has found that anticipation of future events is a major driver of sports-viewing behaviour. Since anticipation has been theorised to be driven by intentions (e.g., Abbate, 2025), it is reasonable to assume that sports-viewing intentions are a driver of behaviour. Future research should seek to assess the extent to which intentions, as measured by the MGB or SFMGB, translate into behaviour in the context of Formula 1. This could be achieved by utilising a longitudinal approach in which subsequent behaviour is measured at a later date.

Conclusion

The present study explored the applicability of the SFMGB in the context of Formula 1, and whether providing viewers with a data dashboard could improve intentions to engage with the sport. It aimed to address the lack of literature surrounding the validity of the SFMGB and was the first to investigate this model in relation to Formula 1. In relation to this aim, the study found limited evidence for the mediation effects proposed by the SFMGB. The model was found to be overwhelmingly predictive of intentions, but this was most likely overinflated due to methodological limitations. It is also likely that the increased quantity of determinants over the MGB have overcomplicated the model and led to increased multicollinearity of the determinants, which diluted their individual effects. Overall, there is insufficient evidence that the SFMGB is applicable to Formula 1. The results of both this study and previous research suggest that the high complexity of the SFMGB does not significantly improve its' explanatory power over the standard MGB and may even compromise its applicability. This study also aimed to investigate the extent to which emerging fan-oriented data dashboards could be provided by Formula 1 to improve engagement. Here, it was found that the data dashboard had a positive impact on engagement intentions regardless of fan status. This suggests that Formula 1 may benefit from offering viewers a dedicated data dashboard to enhance the viewer experience and increase the likelihood of future engagement. However, it is unclear how this could be used to encourage non-fans or prospective viewers to engage. Furthermore, methodological limitations limit the reliability of these results, and conceptual limitations such as the Intention-Behaviour Gap restrict the extent to which these results can explain real behaviour. Overall, this study highlights the need for further research to fully determine the potential impact of an official Formula 1 data dashboard on engagement with the sport. Nevertheless, the findings largely align with existing literature which suggests that data dashboards can facilitate engagement. The limitations of this research also point towards the need for a more comprehensive assessment of the SFMGB with a larger sample size, more robust measures of participants' engagement with the survey contents, and measures of real-life sports-engagement behaviours.

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Appendix A

Questionnaire Items

(1) “I think watching Formula 1 is…”

Bad 1 - 2 - 3 - 4 - 5 - 6 - 7 Good

(2) “I think watching Formula 1 is…”

Harmful 1 - 2 - 3 - 4 - 5 - 6 - 7 Beneficial

(3) “I think watching Formula 1 is…”

Unpleasant 1 - 2 - 3 - 4 - 5 - 6 - 7 Pleasant

(4) "People who are important to me think I (Should/Should Not) watch Formula 1."

Should 1 - 2 - 3 - 4 - 5 - 6 - 7 Should Not

(5) "People who are important to me would (Approve/Disapprove) of me watching Formula 1."

Approve 1 - 2 - 3 - 4 - 5 - 6 - 7 Disapprove

(6) How much control do you have over whether you watch Formula 1?

No Control 1 - 2 - 3 - 4 - 5 - 6 - 7 Total Control

(7) "If I wanted to watch F1, it would be..."

Difficult 1 - 2 - 3 - 4 - 5 - 6 - 7 Easy

(8) If you watched the next Formula 1 race, how would you feel?

Happy 1 - 2 - 3 - 4 - 5 - 6 - 7 Unhappy

(9) If you watched the next Formula 1 race, how would you feel?

Excited 1 - 2 - 3 - 4 - 5 - 6 - 7 Bored

(10) If you watched the next Formula 1 race, how would you feel?

Satisfied 1 - 2 - 3 - 4 - 5 - 6 - 7 Disappointed

(11) If you missed the next F1 race, how would you feel?

Frustrated 1 - 2 - 3 - 4 - 5 - 6 - 7 Indifferent

(12) If you missed the next F1 race, how would you feel?

Regretful 1 - 2 - 3 - 4 - 5 - 6 - 7 Unconcerned

(13) If you missed the next F1 race, how would you feel?

Angry 1 - 2 - 3 - 4 - 5 - 6 - 7 Calm

(14) If you missed the next F1 race, how would you feel?

Uncomfortable 1 - 2 - 3 - 4 - 5 - 6 - 7 Comfortable

(15) "I want to watch Formula 1"

False 1 - 2 - 3 - 4 - 5 - 6 - 7 - 8 - 9 - 10 - 11 True

(16) "I desire to watch Formula 1"

False 1 - 2 - 3 - 4 - 5 - 6 - 7 - 8 - 9 - 10 - 11 True

(17) "My desire to watch Formula 1 is…"

Non-Existent – Very Weak – Weak – Moderate – Strong – Very Strong

(18) "I frequently talk about Formula 1 with others"

Strongly Disagree – Disagree – Somewhat Disagree – Neither Agree Nor Disagree – Somewhat Agree – Agree – Strongly Agree

(19) "I interact with Formula 1 content on Social Media"

Strongly Disagree – Disagree – Somewhat Disagree – Neither Agree Nor Disagree – Somewhat Agree – Agree – Strongly Agree

(20) "I advise or encourage others to watch Formula 1"

Strongly Disagree – Disagree – Somewhat Disagree – Neither Agree Nor Disagree – Somewhat Agree – Agree – Strongly Agree

(21) "I engage in online discussions about Formula 1"

Strongly Disagree – Disagree – Somewhat Disagree – Neither Agree Nor Disagree – Somewhat Agree – Agree – Strongly Agree

(22) "I identify with the Formula 1 fan community"

Strongly Disagree –Somewhat Disagree – Neither Agree Nor Disagree – Somewhat Agree – Strongly Agree

(23) "My self-image overlaps with the identity of the Formula 1 fan community”

Strongly Disagree –Somewhat Disagree – Neither Agree Nor Disagree – Somewhat Agree – Strongly Agree

(24) "I consider myself a supporter of a particular Formula 1 team"

Strongly Disagree – Disagree – Somewhat Disagree – Neither Agree Nor Disagree – Somewhat Agree – Agree – Strongly Agree

(25) "I feel a strong sense of loyalty to a particular Formula 1 team"

Strongly Disagree – Disagree – Somewhat Disagree – Neither Agree Nor Disagree – Somewhat Agree – Agree – Strongly Agree

(26) "I am proud to be a fan of my favourite Formula 1 team"

Strongly Disagree – Disagree – Somewhat Disagree – Neither Agree Nor Disagree – Somewhat Agree – Agree – Strongly Agree

(27) "Thinking about the Formula 1 video I was shown, I was satisfied with it"

Strongly Disagree – Disagree – Somewhat Disagree – Neither Agree Nor Disagree – Somewhat Agree – Agree – Strongly Agree

(28) "Watching the Formula 1 video was enjoyable"

Strongly Disagree – Disagree – Somewhat Disagree – Neither Agree Nor Disagree – Somewhat Agree – Agree – Strongly Agree

(29) "I intend to watch Formula 1 again in the future"

Strongly Disagree – Disagree – Somewhat Disagree – Neither Agree Nor Disagree – Somewhat Agree – Agree – Strongly Agree

(30) "I intend to engage with Formula 1 content on social media"

Strongly Disagree – Disagree – Somewhat Disagree – Neither Agree Nor Disagree – Somewhat Agree – Agree – Strongly Agree

(31) "I intend to follow Formula 1 news and updates online"

Strongly Disagree – Disagree – Somewhat Disagree – Neither Agree Nor Disagree – Somewhat Agree – Agree – Strongly Agree

Ethical Approval Sheet

Downloaded: 20/08/2025
Approved: 19/05/2025

Sam Kensett
Registration number: 240175348
Information School [a.k.a iSchool]

Dear Sam

PROJECT TITLE: Does the use of a data dashboard impact intentions to engage with F1?
APPLICATION: Reference Number 067995
ORIGINAL APPLICATION: Reference Number 067076

On behalf of the University ethics reviewers who reviewed your project, I am pleased to inform you that on 19/05/2025 the above-named project was **approved** on ethics grounds, on the basis that you will adhere to the following documentation that you submitted for ethics review:

- University research ethics application form 067995 (form submission date: 15/05/2025); (expected project end date: 27/08/2025). This is an en bloc application based on University research ethics application form 067076
- Participant information sheet 1152807 version 1 (07/05/2025).
- Participant consent form 1152808 version 1 (07/05/2025).

If during the course of the project you need to deviate significantly from the above-approved documentation please inform me since written approval will be required.

Your responsibilities in delivering this research project are set out at the end of this letter.

Yours sincerely

Nia Hunt
Ethics Admin
Information School [a.k.a iSchool]

Please note the following responsibilities of the researcher in delivering the research project:

- The project must abide by the University's Research Ethics Policy: https://www.sheffield.ac.uk/research-services/ethics-integrity/policy
- The project must abide by the University's Good Research & Innovation Practices Policy: https://www.sheffield.ac.uk/polopoly_fs/1.671066!/file/GRIPPolicy.pdf
- The researcher must inform their supervisor (in the case of a student) or Ethics Admin (in the case of a member of staff) of any significant changes to the project or the approved documentation.
- The researcher must comply with the requirements of the law and relevant guidelines relating to security and confidentiality of personal data.
- The researcher is responsible for effectively managing the data collected both during and after the end of the project in line with best practice, and any relevant legislative, regulatory or contractual requirements.

Dissertation Research Diary

Meeting 1 Date: 02/05/2025

Work completed:

- Proposal feedback actioned
- Questionnaire constructed

Meeting contents

In this meeting we discussed the feedback on my dissertation proposal, particularly focusing on the structure of my literature review and how this differs from my background in Psychology, as well as considering how my research approach could be expanded to included ontology and epistemology. We then clarified where I feel I am in my schedule and what I intend to focus on in coming weeks. Finally, we clarified certain methodological aspect of the study such as the questionnaire and the data dashboard. Dave was satisfied that these would work for my study, however he identified participant recruitment as a potential obstacle. I intend to consider ways in which I can maximise participant recruitment.

Planned goals

- Information sheet completion
- Consent form completion
- Ethics application
- Consideration of participant recruitment

Meeting 2 Date: 18/06/2025

Work completed

- Ethics approval received
- Questionnaire sent out

Meeting contents

We discussed further methods of participant recruitment, including the volunteer list. We also clarified some of the feedback from the dissertation proposal. I will be focusing on the methodology, introduction and literature review sections over the coming month while I continue to receive questionnaire responses. By the next meeting, I intend to have finished collecting responses and may have begun analysis in SPSS.

Planned goals

- Complete methodology, introduction and literature review sections.
- Finish taking questionnaire responses
- Begin data analysis

Meeting 3 Date: 14/07/2025

Work completed

- Questionnaire responses complete
- Data analysis started
- Introduction, literature review, and methodology feedback actioned.

Meeting contents

We discussed the start of data analysis and the plan for constructing the model through partial mediation. I identified an outcome from the current data related to a RQ, despite the challenges in participant recruitment.

We discussed how the issue of participant recruitment can be considered in the dissertation and reflected upon for the design and planning of studies like this and what it means for the interpretation of results.

Planned goals

- Complete data analysis
- Complete results section
- Begin discussion section

Meeting 4 Date: 08/08/2025

Work completed

- Data analysis complete
- Results section complete
- Discussion section complete

Meeting contents

In this meeting we discussed the feedback for the analysis and discussion sections, with only minor amendments and considerations needed. No concerns were raised, I am on target to finish my dissertation well in time for the deadline. We also set out the work for this month which includes the research diary, reflection, and the structured abstract.

Planned goals

- Complete research diary and reflection
- Amend small formatting anomalies
- Complete structured abstract

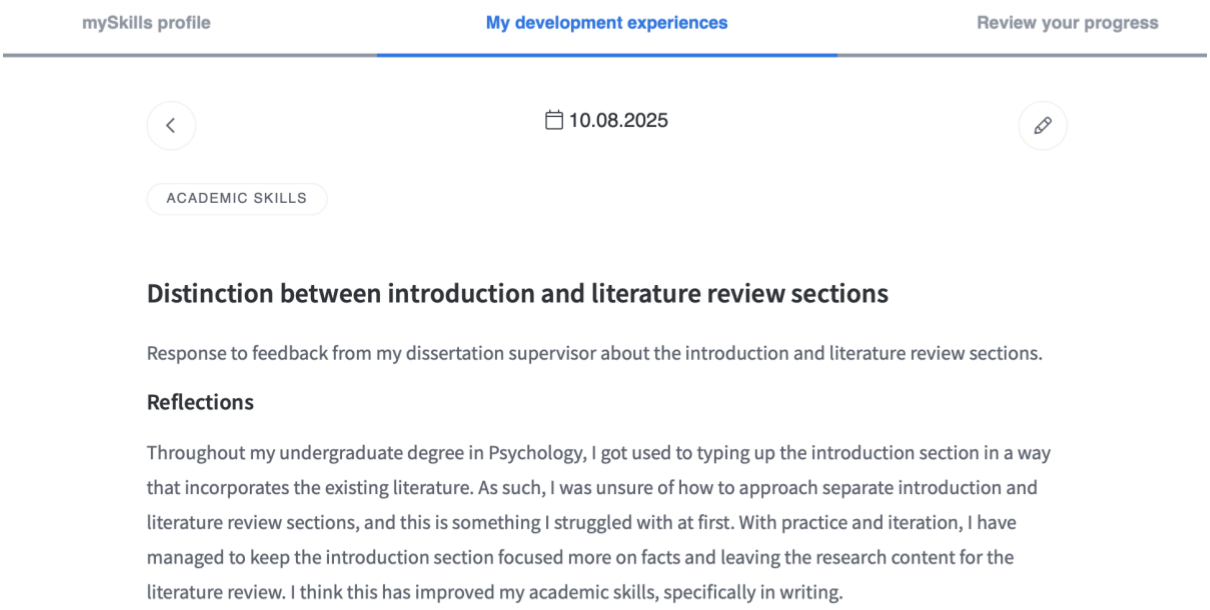
Reflection on Research

Supervisor Feedback 1

Feedback: Create more of a clear distinction between the introduction and literature review sections.

Reflection on feedback: Throughout my undergraduate degree in Psychology, I got used to typing up the introduction section in a way that incorporates the existing literature. As such, I was unsure of how to approach separate introduction and literature review sections, and this is something I struggled with at first. With practice and iteration, I have managed to keep the introduction section focused more on facts and leaving the research content for the literature review. I think this has improved my academic skills, specifically in writing.

Associated skill: Academic Skills



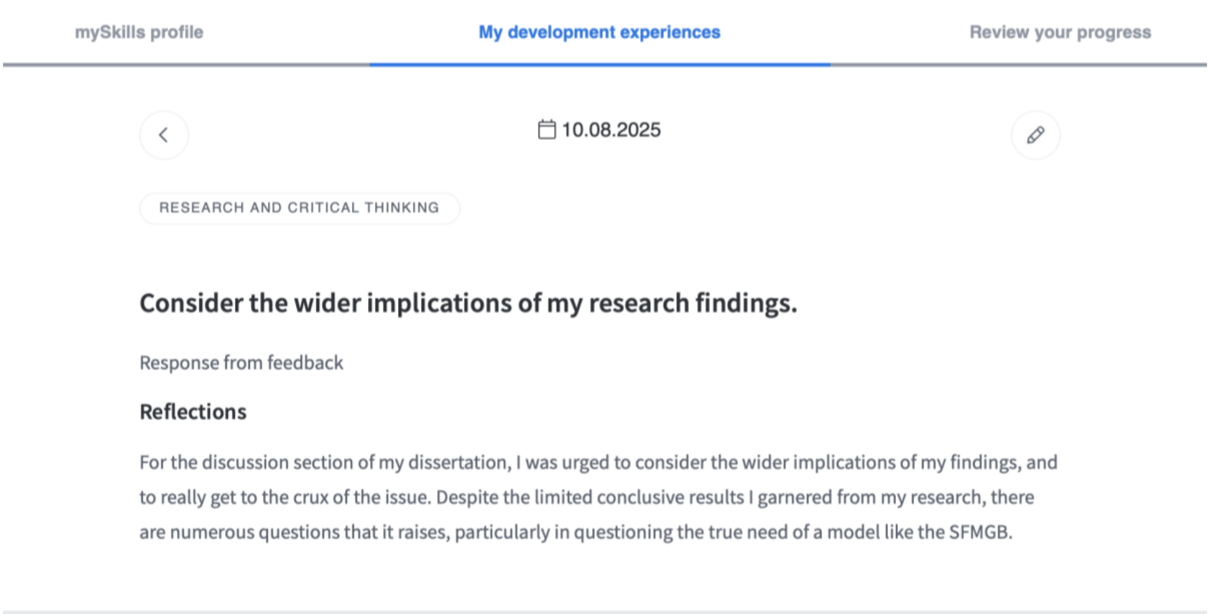
Supervisor Feedback 2

Feedback: Consider the wider implications of your research findings.

Reflection on feedback: For the discussion section of my dissertation, I was urged to consider the wider implications of my findings, and to really get to the crux of the issue. Despite the limited conclusive results I garnered from my research, there are numerous questions that it raises, particularly in questioning the true need of a model like the SFMGB.

Associated skill: Research and Critical Thinking

mySkills Screengrab



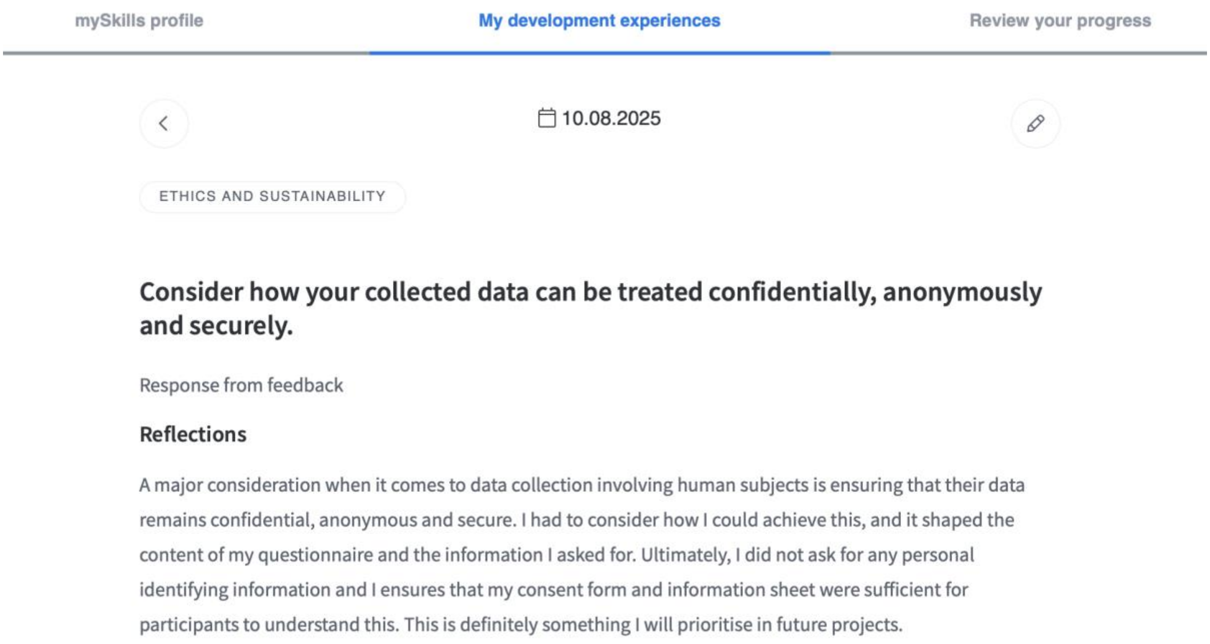
Supervisor Feedback 3

Feedback: Consider how your collected data can be treated confidentially, anonymously and securely.

Reflection on feedback: A major consideration when it comes to data collection involving human subjects is ensuring that their data remains confidential, anonymous and secure. I had to consider how I could achieve this, and it shaped the content of my questionnaire and the information I asked for. Ultimately, I did not ask for any personal identifying information and I ensures that my consent form and information sheet were sufficient for participants to understand this. This is something I will prioritise in future projects.

Associated skill: Ethics and Sustainability

mySkills Screengrab



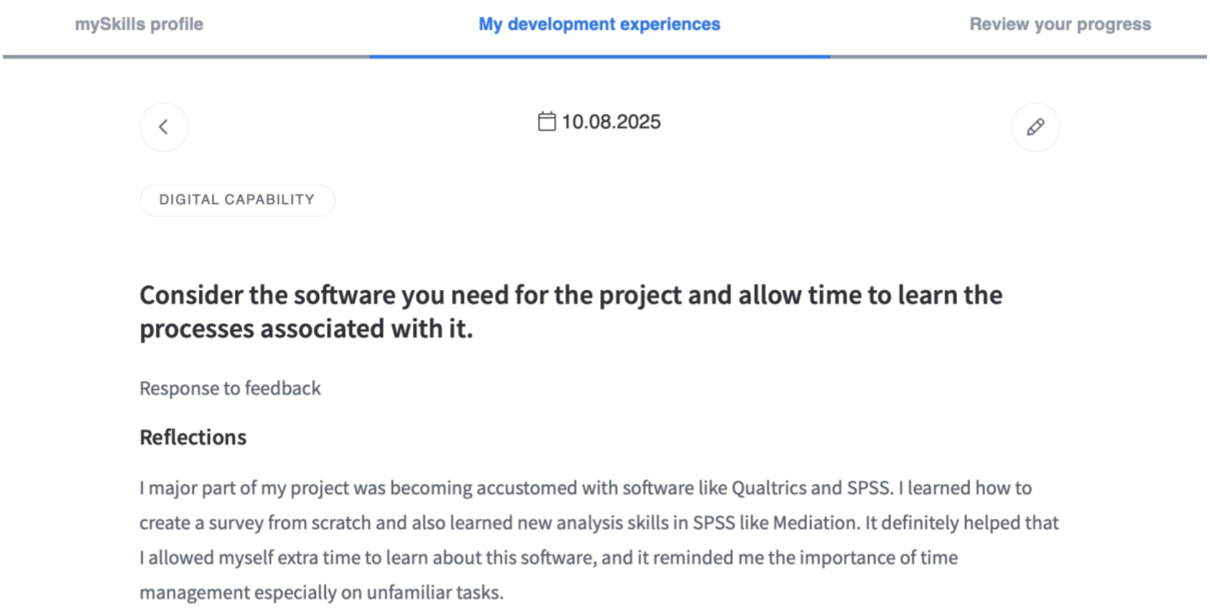
Supervisor Feedback 4

Feedback: Consider the software you need for the project and allow time to learn the processes associated with it.

Reflection on feedback: I major part of my project was becoming accustomed with software like Qualtrics and SPSS. I learned how to create a survey from scratch and learned new analysis skills in SPSS like Mediation. It helped that I allowed myself extra time to learn about this software, and it reminded me the importance of time management especially on unfamiliar tasks.

Associated skill: Digital Capability

mySkills Screengrab



Final Note

Working through this dissertation has supported my personal development.

Working through this project - from the formation of research questions and methodology design, all the way to data collection, findings and critical evaluation – has been a test of my resilience and commitment. Even in the face of challenges like developing a survey people might want to take part in, and understanding a research area which is currently scarce, I have found it extremely rewarding and the skills I have developed will serve me long into the future.

Associated skill: Personal Development

mySkills Screengrab